SECP2613 SYSTEM ANALYSIS & DESIGN
ASSIGNMENT 2: PROJECT MANAGEMENT

SECTION 02
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QUESTION 1

A. Technologically, it’s feasible as the technology to update the system is available. However, operationally, the current system is functioning well and staff are satisfied with it. Implementing a new system could disrupt operations. Economically, the costs of implementing a new system, including potential disruption to business operations during the transition period, need to be considered. The economic benefits of the new system are unclear if the current system is satisfied by staff and functioning well. So, while the project is technologically feasible, its operational and economic feasibility are unclear. A more detailed feasibility study considering all aspects would provide a more comprehensive assessment.

B. Based on Abu’s feedback from the managers, users, and systems people, the operational feasibility of the proposed project is low. The current system is functioning well and the staff is happy and satisfied.

C. Implementing a new system would likely involve significant costs, including the purchase of the new hardware, and software, and hiring and training staff to operate and maintain the new system. While others are so satisfied with it, the economic benefits of the new system are unclear. However, a detailed economic feasibility study considering all costs and potential benefits would provide a more accurate assessment.

D. The technological feasibility seems to be high. From the conversation, the technology to update the system is available. However, technological feasibility does not only consider the availability of technology but also whether the technical expertise or staff can maintain it or not. If the staff and others are satisfied with the current system, this could impact the technological feasibility.

E. Based on the discussion between Ahmad and Abu, it seems that a full-blown systems study may not be necessary at this time. Abu pointed out that the current system still functions very well and the staff, managers, and users are satisfied with it. Before committing to a full-blown systems study, it might be more beneficial to explore less cost and less disruptive ways to update the company's image. They can improve marketing strategies or enhance customer services. However, if these efforts do not have any impact, a more detailed systems study should be considered in the future.

QUESTION 2

1. Research Phase:

    1.1 Define Requirements(Duration: 1 week)

    1.2 Researching Market(Duration: 2 weeks)

2. Decision Phase:

    2.1 Options Evaluation(Duration: 2 weeks)

    2.2 Select Vendor/Develop Team(Duration: 1 week)

3. Development Phase:

    3.1 Design System Architecture(Duration: 3 weeks)

    3.2 Develop Registration System(Duration: 6 weeks)

4. Testing Phase:

    4.1 Conduct Internal Testing(Duration: 2 weeks)

    4.2 Gather User Feedback(Duration: 1 week)

5. Deployment Phase:

    5.1 Staff Training(Duration: 1 week)

    5.2 Rollout System(Duration: 1 week)

Research referred to :

N. Y. Lee, "Conceptual WBS model for a Large-Scale Information System Acquisition Project," 2001. https://koreascience.kr/article/JAKO200111922691159.page

QUESTION 3

Title: The Distinction Between Agile and Traditional Project Management

Introduction: Project management has two primary methodologies, namely traditional and agile. The traditional approach, exemplified by the waterfall model, adheres to a linear and sequential process. In contrast, agile methods such as Scrum emphasize adaptability, flexibility, and incremental progress.

Key Differences:

-Adaptability In traditional project management, a clear roadmap is laid out to reach the project's goals, and any alterations require managerial approval. On the other hand, Agile methodology accommodates shifting priorities throughout the project lifecycle, allowing for modifications at any stage.

-Tracking Progress In a traditional setup, the project manager liaises with stakeholders and conducts weekly team meetings. Agile teams, however, hold brief daily meetings to pinpoint issues, keeping all stakeholders abreast of the project's status.

-Documentation While traditional project management mandates comprehensive documentation at each phase, Agile focuses on operational systems, processes, and regular customer feedback. The project manager and team members record their analyses, decisions, and outcomes at every stage.

-Testing and Validation In the traditional model, dedicated teams conduct testing towards the end of the project, whether it's software or a new product. In contrast, Agile involves frequent testing by users and customers at each stage.

Conclusion: In summary, Agile project management emphasizes adaptability, robust stakeholder engagement, and the delivery of incremental value. Traditional project management, on the other hand, follows a set timeline, predictability, and thorough documentation. The choice between these two methodologies depends on the specific goals and objectives of the project.

QUESTION 4

A.

Costs	Year 0	Year 1	Year 2	Year 3
Development Costs				
Hardware	RM50,000			
Software	RM15,000			
Training	RM30,000			
Consulting	RM80,000			
Data Conversion	RM40,000			
Total	RM215,000			
Production Costs				
Supplies		RM15,000	RM15,000	RM15,000
IS Salaries (10% annual increase)		RM75,000	RM82,500	RM90,750
Upgrades		RM10,000	RM10,000	RM10,000
Annual Prod. Costs		RM100,000	RM107,500	RM115,750
(Present Value)		RM74,074	RM58,985	RM47,046
Accumulated Costs		RM289,074	RM348,059	RM395,105
Benefits				
Improve Customer Service (20%)		RM150,000	RM180,000	RM216,000
Increase productivity (25%)		RM150,000	RM187,500	RM234,375
Annual Benefits		RM300,000	RM367,500	RM450,375
(Present Value)		RM222,222	RM201,646	RM183,051
Accumulated Benefits		RM222,222	RM423,868	RM606,919
Gain or Loss		-RM66,852	RM75,809	RM211,814
Profitability Index	0.985			

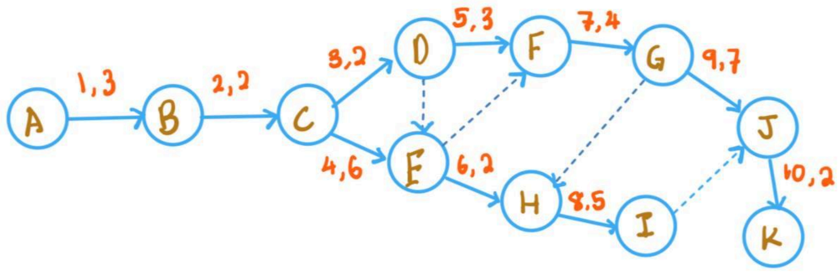
B. Profitability Index = 211814 / 215000

= 0.985

The profitability index value shows that it is not a good investment because its index is less than one. The project's costs outweigh its expected benefits over the 3 years. Hence, investing in the project is not recommended as it is not economically feasible.

QUESTION 5

A.



B. Path 1 = A,B,C,D,F,G,J,K (23 weeks)

Path 2 = A,B,C,E,H,I,J,K (20 weeks)

Path 3 = A,B,C,D,E,F,G,H,I,J,K (18 weeks)

Path 4 = A,B,C,E,F,G,H,I,J,K (22 weeks)

Path 5 = A,B,C,E,F,G,J,K (24 weeks)

C. Critical Path is the longest sequence of dependent tasks in a project therefore Path 5 which isA,B,C,E,F,G,J,K with a duration of 24 weeks is the critical path.

D.